Year 8
Mathematics

Unit 2: Equations and Formulae

2.2 Solving Two-Step Equations

Solving Two-Step Equations

STARTER · 5 MIN

Date: _____ **Title: Solving Two-Step Equations**

LEARNING OBJECTIVES

- To be able to write and solve two-step equations

STARTER *Answer in your book. No calculators.*

1. Solve $x + 5 = 11$

2. Solve $3x = 21$

3. Solve $y - 4 = 10$

4. $\frac{m}{2} = 9$. Find m.

Starter — Answers

STARTER

Tick or correct in green pen

1. $x = 6$

2. $x = 7$

3. $y = 14$

4. $m = 18$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use inverse operations to solve two-step equations	☐
SILVER	I can form and solve more complex equations	☐
GOLD	I can write a two-step equation from a word problem and solve it correctly	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Solving Two-Step Equations

INTRODUCTION · 3 MIN

A two-step equation needs two inverse operations. Undo the + or − first, then the × or ÷.

Two steps

The letter has two operations.

$2x + 3 = 9$

Step 1

Undo + or −.

$2x = 9 - 3 = 6$

Step 2

Undo × or ÷.

$x = 6 \div 2 = 3$

Check

Substitute back in.

$2 \times 3 + 3 = 9$, correct

Examples 1 and 2 — Two Inverse Operations

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Undo + or − first Then undo × or ÷

Example 1

Solve

$2x + 3 = 9$

Example 2

Solve

$3x - 4 = 8$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Subtract 3, then divide by 2

$2x + 3 = 9$

$2x = 6$

$\boldsymbol{x = 3}$

Tip: undo + 3 first.

Example 2

Add 4, then divide by 3

$3x - 4 = 8$

$3x = 12$

$\boldsymbol{x = 4}$

Tip: check: 3 × 4 − 4 = 8.

Task 1

PRACTICE · 9 MIN

BRONZE *Solve*

Hint Undo + or − first Then undo × or ÷

1. $2v + 1 = 23$

2. $5x - 2 = 18$

3. $3h + 4 = 13$

4. $4x - 9 = 15$

5. $5x + 7 = 32$

★ $2a + 1 = 10$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $v = 11$

2. $x = 4$

3. $h = 3$

4. $x = 6$

5. $x = 5$

★ $a = 4.5$

Examples 3 and 4 — Fractions and Brackets

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Bracket: divide first Fraction: × last

Example 3

Solve

$$\frac{w}{4} - 5 = 15$$

Example 4

Solve

$$3(u + 2) = 21$$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Add 5, then multiply by 4

$$\frac{w}{4} - 5 = 15$$

$$\frac{w}{4} = 20$$

$$\boldsymbol{w = 80}$$

Tip: undo − 5 first.

Example 4

Divide by 3, then subtract 2

$$3(u + 2) = 21$$

$$u + 2 = 7$$

$$\boldsymbol{u = 5}$$

Tip: or expand first: $3u + 6 = 21$.

Task 2

PRACTICE · 9 MIN

SILVER *Solve*

Hint Bracket: divide first
Fraction: × last

1. $\frac{x}{3} + 2 = 7$

2. $\frac{y}{5} - 1 = 4$

3. $2(x + 3) = 14$

4. $4(m - 1) = 20$

5. $\frac{a + 4}{2} = 6$

★ $5(2t - 1) = 35$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $x = 15$

2 $y = 25$

3 $x = 4$

4 $m = 6$

5 $a = 8$

★ $t = 4$

Examples 5 and 6 — Writing Two-Step Equations

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Write the equation first Check by substituting

Example 5

(2 marks)

I think of a number, × 4, then subtract 7. The answer is 25.

Find my number.

Example 6

(3 marks)

A taxi costs $3 plus $2 per km. A trip costs $19.

How many km was it?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Write the equation, then solve

$4n - 7 = 25$

$4n = 32$

$\boldsymbol{n = 8}$

Tip: check: $4 \times 8 - 7 = 25$.

Example 6

Fixed charge + rate × km

$3 + 2k = 19$

$2k = 16$

$\boldsymbol{k = 8}$ **km**

Tip: say what k stands for.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: write and solve*

Hint Write the equation first Check by substituting

1. Double a number, add 5: the answer is 21. Find it.

2. Gym: $20 + $15 a month. Cost $110. How many months?

3. Triangle angles: x, $2x$ and 90°. Find x.

4. Rectangle x cm by 5 cm. Perimeter 26 cm. Find x.

5. 3 pens at p cents and a 50c ruler cost 260c. Find p.

★ Half a number, minus 3, is 9. Find the number.

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $2n + 5 = 21$: $n = 8$
2. $20 + 15m = 110$: $m = 6$
3. $3x + 90 = 180$: $x = 30°$
4. $2x + 10 = 26$: $x = 8$
5. $3p + 50 = 260$: $p = 70$

★ $\frac{n}{2} - 3 = 9$: $n = 24$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Solve $2x + 3 = 9$

SILVER Solve $4x - 9 = 15$

GOLD Solve $5x + 7 = 32$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $x = 3$

SILVER $x = 6$

GOLD $x = 5$

Exam Practice — Two-Step Equations

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Undo + or − first Then undo × or ÷

Question 1

(2 marks)

Solve $3x + 5 = 26$

$x =$ .....

Question 2

(2 marks)

Solve $6y - 8 = 34$

$y =$ .....

Exam Practice — Brackets and Fractions

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Bracket: divide first Fraction: × last

Question 3

(2 marks)

Solve $4(x - 3) = 28$

$x =$ .....

Question 4

(2 marks)

Solve

$$\frac{m}{3} + 4 = 10$$

$m =$ .....

Exam Practice — Equations from Words

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Write the equation first
Check by substituting

Question 5

(3 marks)

The angles of a triangle are x, $x + 20$ and $2x$.

Write an equation and find x.

..... °

Question 6

(3 marks)

A plumber charges $40 plus $25 per hour. A job costs $165.

How many hours did the job take?

..... hours

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$3x = 21$ — M1

$\boldsymbol{x = 7}$ — A1

Tip: − 5, then ÷ 3.

Question 2

$6y = 42$ — M1

$\boldsymbol{y = 7}$ — A1

Tip: + 8, then ÷ 6.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$x - 3 = 7$ **M1**

$\mathbf{x = 10}$ **A1**

Tip: or $4x - 12 = 28$.

Question 4

$\frac{m}{3} = 6$ **M1**

$\mathbf{m = 18}$ **A1**

Tip: − 4, then × 3.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$x + x + 20 + 2x = 180$ — M1

$4x = 160$ — M1

$\boldsymbol{x = 40°}$ — A1

Tip: angles in a triangle add to 180°.

Question 6

$40 + 25h = 165$ — M1

$25h = 125$ — M1

$\boldsymbol{h = 5}$ — A1

Tip: undo the fixed charge first.